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SmartTD: A Web Platform for Tie-Dye Pattern Generation and Digital Innovation

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***Abstract*—Today, the utilization of modern technological means to advance traditional tie-dye techniques has emerged as a critical issue that requires immediate attention. To address this, we propose an approach that integrates information technology with traditional craftsmanship through a web platform known as SmartTD. SmartTD efficiently extracts and analyzes existing tie-dye patterns to generate new and creative tie-dye artworks. This tie-dye pattern generation not only significantly expands users' creative options but also infuses new vitality into traditional handicrafts. In addition, SmartTD retraces the historical background of tie-dye and its unique development trajectories in different regions of China, providing abundant technical documentation and details. Users can simulate the actual tie-dye process through virtual experiences offered on SmartTD, thereby gaining a more comprehensive understanding of the cultural value of this ancient craft. The integrated application of these methodologies provides a novel practical solution for the digital preservation of traditional handicrafts while simultaneously opening up broader avenues for cultural education and dissemination.**

***Index Terms*—GenAI, tie-dye, traditional craftsmanship, digitization.**

I. Introduction

Today, traditional craftsmanship faces unprecedented challenges and opportunities, particularly within the realm of intangible cultural heritage (ICH). The application of digital technology to promote the innovative development of traditional crafts has become an urgent topic [1]. Tie-dye, a craft with a long history and rich cultural connotations, was inscribed on the Chinese National List of Intangible Cultural Heritage in 2006, representing a valuable crystallization of folk art. However, the inheritance of tie-dye techniques is faced with numerous difficulties. Firstly, there is a challenge in passing on ICH due to a waning interest among younger generations in traditional crafts. Secondly, the aging of inheritors leads to a diminishing number of tie-dye artisans. Furthermore, the lack of innovation in traditional tie-dye designs makes it difficult to attract new consumers, resulting in a shortage of innovative talent and forming a vicious cycle [2].

To protect ICH and expand its dissemination, digital methods have become essential. With the continuous evolution and improvement of generative artificial intelligence (GenAI) technologies, GenAI images are increasingly entering public awareness. Given the characteristics of GenAI imagery, such as rapid production, low cost, and customizability, it has penetrated various fields. Applying GenAI to the field of tie-dye can potentially alleviate issues related to the high costs associated with tie-dye pattern production and the difficulty of digitizing these patterns. We propose an online platform to experience the Chinese tie-dye culture named SmartTD. This platform showcases the historical development of tie-dye, provides an experience of the crafting process, and offers users a preliminary understanding of the technique. Subsequently, users can upload their desired tie-dye patterns or describe their envisioned styles, upon which the platform will generate unique tie-dye designs based on user input. The platform leverages GenAI to create tie-dye patterns. By learning from a vast repository of historical tie-dye designs, the GenAI model can autonomously generate novel designs. Through the integration of GenAI with the ICH tie-dye, SmartTD aims to inspire design creativity and contribute to the preservation and promotion of tie-dye craftsmanship.

The main contributions of this paper are summarized as follows.

- We combine GenAI with tie-dye craftsmanship to develop a platform for digital preservation and promotion of the traditional art form while making it accessible to a wider audience.
- The platform offers users with an educational and entertaining experience about the historical development of tie-dye in China, providing technical simulation insights and improving cultural understanding of this ancient craftsmanship.
- We collect a series of traditional tie-dye images and use these images as a dataset to fine-tune the Stable Diffusion model with the LoRA method, achieving better control over style transfer. Then, the platform are able to generate unique, non-repetitive patterns with distinctive tie-dye characteristics, further enriching the digital design innovations in tie-dye craftsmanship.

II. Related work

Currently, existing digital platforms dedicated to ICH predominantly offer informational and encyclopedic multimedia resources for public access. Some platforms that incorporate sections on traditional craftsmanship provide rudimentary interactive experiences to illustrate the fabrication processes. However, these platforms offer limited opportunities for user creation and engagement, preventing users from experiencing the creation of artworks in the relevant artistic style within

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an ICH-themed environment. Furthermore, specialized digital platforms focusing on tie-dye as an ICH element are scarce in the current. Despite this scarcity, many studies have explored the feasibility of integrating tie-dye with information technology. Zheng et al. [3] utilized the Processing sound visualization technology to programmatically express abstract elements, combining audio input to generate four different styles of tie-dye patterns. This method simulates the dyeing effect of tie-dye to produce patterns, offering new inspiration and ideas for pattern design in fashion and interactive design. Dou et al. [4], based on Mandelbrot set graphics, proposed a design method for tie-dye-like patterns. By adjusting the parameters and functions of the Mandelbrot mathematical model, they generated varying styles of Mandelbrot graphic transformations through progressive magnification and color-mode adjustments, applying the designs to silk scarves to achieve tie-dye-like effects. Similarly, in the context of Fengyang Phoenix Painting, a form of ICH, Zhang et al. [5] used target detection and style transfer algorithms within GenAI technology to extract and collect characteristic features and styles of Fengyang Phoenix Painting, contributing to their digital preservation and creative design, integrated into a Fengyang Phoenix Painting creation platform.

These findings indicate that the application of information technology and GenAI could broaden the channels of dissemination of traditional Chinese ICH, achieving cultural innovation through the fusion of tradition and technology.

III. Model and Dataset

This study initially collected a series of traditional tie-dye images from online sources and field visits to tie-dye studios in Guizhou, ensuring that these images aligned with the desired aesthetic. To date, we have collected a dataset comprising 528 tie-dye images. Given the constraints imposed by our limited number of tie-dye samples, we have opted to employ the Low-Rank Adaptation (LoRA) method for fine-tuning the Stable Diffusion (SD) model. Since SD model demonstrates superior stability and adaptability compared to a generative adversarial network (GAN), along with improved controllability in style transfer [6]. The LoRA excels in its ability to update only a minimal set of additional parameters within the SD model. Consequently, LoRA makes the training process more efficient, leading to a substantial reduction in the computational resources required.

This study focuses on traditional tie-dye patterns, which typically feature indigo blue as the primary color, producing a gradient effect. The resulting patterns vary depending on the binding locations and techniques employed. To facilitate annotation of images in the dataset, the collected tie-dye patterns were categorized into several categories.

1) Continuous patterns: These include designs such as fish-scale patterns, linear patterns, and radial shapes that are typically continuous in nature.
2) Symmetrical patterns: Circular, fan-shaped, and polygonal patterns that are commonly found centered and symmetrical in traditional tie-dye designs.
3) Isolated patterns: During field visits, many patterns derived from modern needs were collected along with traditional ones, including motifs like butterflies and flowers, which are often singular in form. These were classified as isolated patterns.
4) Composite patterns: These represent more complex contemporary designs that integrate elements of isolated patterns with symmetrical or continuous motifs, potentially incorporating intricate layouts and designs.

As an illustration, Figure 1 provides examples of the four pattern types mentioned above.

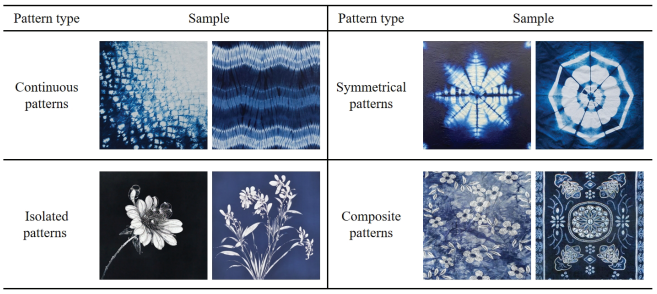


Fig. 1: Classification table of tie-dye patterns.

Following classification, then tag the dataset as tie-dye style. Subsequently, manual annotations were added according to the aforementioned pattern classifications, providing detailed descriptions of the patterns. Once annotated, ChilloutMix, chosen for its realistic appearance, was adopted as the base model for adjustments. The dataset underwent fifteen rounds of training, resulting in a LoRA model that exhibits distinct tie-dye characteristics.

Consequently, based on the model obtained from the aforementioned training, users can input textual descriptions of desired elements as prompts to generate unique, non-replicable patterns. In addition, users are also able to upload any preferred images to receive a version imbued with the distinctive characteristics of the tie-dye style. In this study, floral, avian, insect, and traditional landscape paintings were selected as four groups for style transfer comparisons, with the results illustrated in Figure 2.

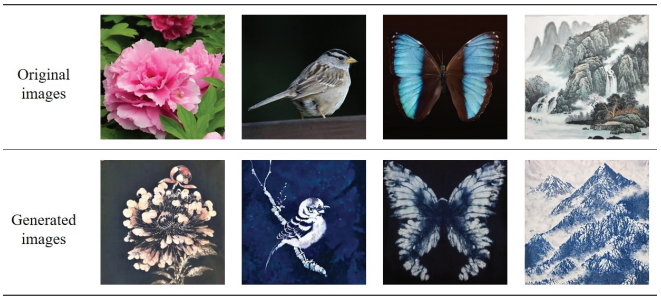


Fig. 2: Visual representations of tie-dye styles.

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IV. Design and User Interface

An online intelligent platform for ICH represents a means to present traditional ICH within the context of the Internet [7], thus freeing ICH dissemination from temporal and spatial constraints. The SmartTD platform, which serves as a comprehensive digital platform that includes knowledge dissemination, experiential learning, entertainment, and creation, not only reduces the time and financial costs associated with engaging in tie-dye craftsmanship but also broadens the reach and awareness of traditional Chinese tie-dye culture, promoting a wider distribution of Chinese ICH.

A. User interface of SmartTD

The user interface design of the SmartTD webpage integrates modern aesthetics with the artistic qualities of tie-dye, as shown in Figure 3. Adhering to contemporary web design standards, the color scheme centers around fresh and natural tones, with indigo blue and off-white, commonly seen in tie-dye, as the primary hues, evoking an atmosphere of rustic elegance. In terms of layout, tie-dye patterns are incorporated ingeniously into various aspects of the webpage, such as backgrounds, icons, and dividers, maintaining a consistent design style throughout. At the level of detail, the webpage employs the unique fabric textures and diffusion effects characteristic of tie-dye, enriching the visual experience for users. Through thoughtful application of color, layout, and detail, the webpage showcases the unique charm of tie-dye art, allowing users to appreciate the essence and depth of Chinese traditional culture during their browsing experience.

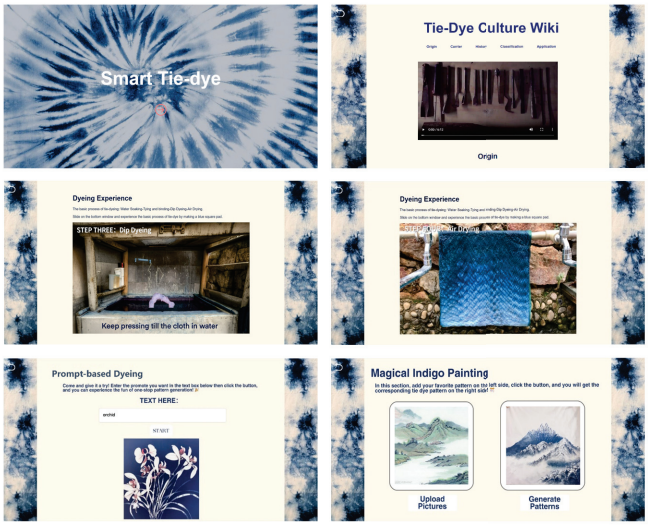


Fig. 3: User interface of SmartTD.

B. Functional design of SmartTD

The functional structure of the SmartTD is divided into three main modules: tie-dye culture wiki, dyeing experience, and intelligent tie-dye design, as depicted in Figure 4. Initially, SmartTD presents a richly illustrated narrative of traditional Chinese tie-dye culture through five categories: origin, carrier, history, classification, and application, to facilitate quick comprehension for new users. Subsequently, SmartTD employs game-like interactions to guide users through the steps and processes of traditional Chinese tie-dye craftsmanship, achieving an educational and entertaining experience. Finally, augmented by GenAI technology, SmartTD incorporates functionalities for designing tie-dye patterns and customizing tie-dye artifacts, thus presenting the entire spectrum of tie-dye culture.

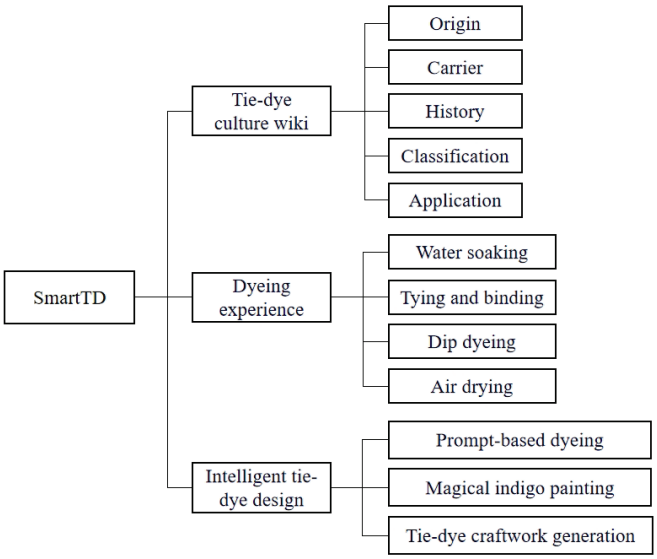


Fig. 4: Functional structure of SmartTD.

1) Tie-dye culture wiki: This module is divided into five main parts: origin, carrier, history, classification, and application. Each part delves into various aspects of the origins of tie-dye patterns, textile culture, historical development, ethnic variations, and applications, providing users with a comprehensive understanding of the evolution of tie-dye art. At the top of this module, a navigation bar allows users to browse sequentially or jump directly to sections of interest. Additionally, the module features an educational video detailing the development of Zigong tie-dye, which auto-plays upon entry but can be paused if users wish to explore other content.

2) Dyeing experience: Within this module, users can interactively experience the complete tie-dye making process. The module guides users through the key steps of "soaking", "tying", "dyeing", and "drying", allowing them to enjoy the pleasure of creating tie-dyes. This module facilitates a deeper understanding of the intricate details of tie-dye craftsmanship, enhancing appreciation for the unique charm of this traditional handcraft. Furthermore, this interactive experience helps stimulate user interest, aiding in the preservation and development of tie-dye skills, blending education with enjoyment.

3) Intelligent tie-dye design: This module offers users three different ways to experience the creation of tie-dye artworks.

Prompt-based dyeing (text-to-image): Users can generate unique tie-dye-style patterns by inputting different prompts.

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Given that the previously trained tie-dye style LoRA model has learned the characteristics of tie-dye patterns, it can automatically convert the textual content into tie-dye-style images that align with the theme described by the text. This text-to-image approach is ideal for users trying the design function for the first time, especially those without specific goals but simply wishing to explore the effects.

Magical indigo painting (image-to-image): Users can upload line arts, photographs, or artworks of any kind, which the model then transforms into artistic pieces imbued with the traditional Chinese tie-dye style and pattern design.

Tie-dye craftwork generation: Users can generate mockups of tie-dye crafts using tie-dye patterns they created on the platform or other tie-dye artwork images. The craft model provided by SmartTD allows users to visualize how their tie-dye patterns would appear on different craft items by simply uploading their images. Beyond traditional textile crafts such as fans, scarves, and canvas bags, users can also experiment with generating crafts made from non-fabric materials such as stickers, bookmarks, or tableware.

The results of each method are illustrated in Figure 5. Users can experience the full spectrum of tie-dye culture, from learning about tie-dye traditions to experiencing the craft process, to creating tie-dye patterns and designing derivative artisanal products, without leaving their homes through SmartTD.

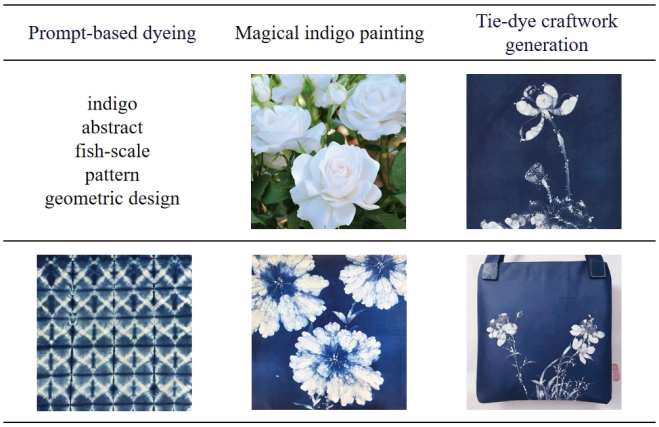


Fig. 5: Intelligent tie-dye design in SmartTD.

V. Evaluation

To assess the public's awareness and attention towards ICH in China and SmartTD, we engaged 103 student volunteers from the School of Media Engineering, Communicaiton University of Zhejiang, to experience the content of SmartTD and participate in a survey. According to the responses to the question on "the level of familiarity with tie-dye techniques," approximately 33% of the respondents indicated that they were "somewhat familiar". More than half of the participants said they were "aware but not familiar" with the tie-dye. Those who were "very familiar" and "not familiar at all" with the tie-dye constituted approximately one-tenth of the total sample size, as illustrated in Figure 6.

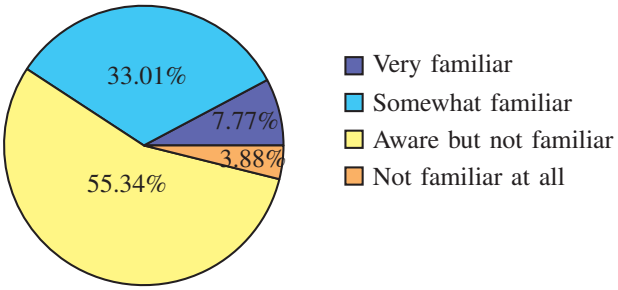


Fig. 6: Public awareness of tie-dye in the survey.

Based on the aforementioned findings, we further investigated the means by which volunteers, excluding those who reported being "not familiar at all", become acquainted with tie-dye techniques in their daily lives. Statistical analysis revealed that almost 80% of the respondents initially became aware of the culture of traditional tie-dye through short videos on the Internet, while only 10% gained understanding through offline hands-on experiences in tie-dye production, as shown in Figure 7.

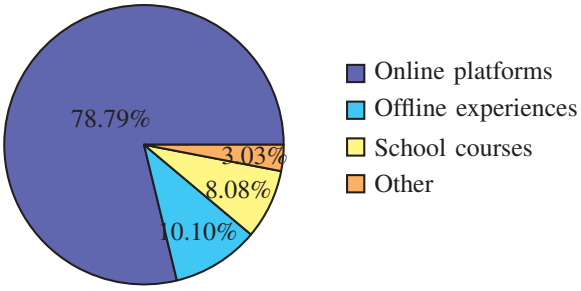


Fig. 7: Ways to aware tie-dye in the survey.

SmartTD is dedicated to broadening the avenues through which people can engage with tie-dye techniques in an era characterized by rapid pace. This initiative aims to enable the public to experience the craft process online and create their own tie-dye artworks, thereby enhancing accessibility and engagement with this traditional art form.

TABLE I: User experience feedback for each module.

	Tie-dye culture wiki	Dyeing experience	Intelligent tie-dye design
Very satisfied	25.24%	25.24%	33.98%
Satisfied	39.81%	47.57%	40.78%
Fair	29.13%	20.39%	10.68%
Unsatisfied	3.88%	4.85%	8.74%
Very unsatisfied	1.94%	1.94%	5.83%

Then, we investigated user experience feedback for each module in SmartTD among 103 student volunteers. As shown in Table I, the "tie-dye culture wiki" module received positive feedback from 67 volunteers who expressed satisfied or very satisfied with the textual and visual materials provided by the user interface. Satisfaction ratings exceeding the "satisfied"

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level accounted for approximately 65% of all evaluations. In general, users praised this module for its comprehensive and concise presentation of tie-dye art, facilitating a deeper understanding of the craft and the rich cultural heritage behind it.

Regarding the "dyeing experience" module, 75 volunteers reported feeling satisfied or very satisfied with the interaction experience, representing 72. 81% of the survey respondents. The proportion of positive feedback was notable. Users praised this section for its clear demonstration process, smooth operational guidance, and the novel nature of the interactive experience, which collectively heightened interest in tie-dye techniques.

Following the utilization of the "intelligent tie-dye design" module, 77 volunteers expressed satisfaction or very satisfaction with the quality of the digitally generated tie-dye patterns, representing nearly 75% of the respondents who completed the questionnaire, with the number of participants selecting "very satisfied" being the highest among all categories. Users noted that the incorporation of GenAI in this module breaks away from conventional design patterns, introducing more personalized and abstract motifs, enabling the creation of tie-dye designs with diverse stylistic characteristics [8]. However, due to the varying needs of different users concerning tie-dye patterns, coupled with the inherent logical limitations present during GenAI image creation, the designs produced did not meet the expectations of some users. This led to an increase in "unsatisfied" feedback within this module.

In summary, the feedback on various SmartTD functionalities has been largely positive, with more than 70% of the volunteers expressing satisfied or higher. Among these, the "Intelligent Tie-dye Design" module garnered the highest number of "very satisfied" responses, while the least number of users rated it as "fair". This outcome indicates that we have effectively met the user's demands for personalized and diversified experiences [9], and provided an innovative visual experience, reflecting the appeal of GenAI-assisted tie-dye creation to the general public. However, the "tie-dye culture wiki" received relatively fewer positive responses compared to other modules, suggesting that a higher standard is expected for traditional text-and-image-based educational content. To address this, we plan to optimize the current educational material and enrich it with multimedia resources such as videos and news updates, ensuring a more comprehensive and engaging knowledge base that better satisfies user needs.

VI. Conclusion

The innovative approach of combining ICH with contemporary technology has become a crucial direction for advancing cultural development. SmartTD emerges in response to this trend, with the objective of rejuvenating traditional tie-dye techniques through information technology means, particularly by leveraging GenAI. Through intelligent and digitalized methods, a greater number of users can gain a deeper understanding and experience of traditional tie-dye techniques, while also being able to create personalized tie-dye artworks in a convenient and cost-effective manner.

Furthermore, we also observed certain issues in SmartTD, such as insufficient pattern collections and virtual experiences that may not be sufficiently realistic, which require further refinement. As an art form steeped in thousands of years of history, the nation is actively seeking breakthroughs to protect and inherit these dyeing techniques. We hope that our design can contribute to efforts to preserve traditional tie-dye crafts.

Acknowledgment

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